
HLYBRO S500 INSPECTION REPORT

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Date : 02 - 07 - 2025
Hanoi, Viet Nam

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1 Configuration

- Frame: Holybro S500
- Motors: Unknown
- Battery: Li-ion 6S 8AH
- Propeller: 9450 Self-tightening propeller
- GNSS: M10 GPS Module
- Flight Controller: Holybro Pixhawk RPi CM4 Baseboard with a Raspberry Pi inside
- Telemetry: SiK Telemetry Radio V3 915MHz
- ESC: Unknown
- Radio Receiver/Controller: Futaba T14SG
- Range Finder/Distance sensor: TFMini Plus

2 Wiring Diagram

Component	Connection Port
GNSS	CAN1
Radio Receiver	I2C
Distance Sensor	TELEM 3
Telemetry Radio	TELEM 1
Motor 1	FMU PWM Output 1
Motor 2	FMU PWM Output 2
Motor 3	FMU PWM Output 3
Motor 4	FMU PWM Output 4

- FMU PWM: This is typically used for controlling the motors and other critical flight control components. It's directly connected to the flight controller's Flight Management Unit (FMU), which handles the main flight operations.
- I/O PWM: This is generally used for non-critical components, such as auxiliary outputs for lights, cameras, or other sensors. It is managed by the Input/Output (I/O) processor, which handles non-flight-related tasks.

3 QGroundControl setup

Airframe: Holybro S500 QuadX Frame

3.1 Sensor Calibration

- Compass: 6 Axis + Rotation
- Gyroscope: Level
- Accelerometer: 6 Axis

- Level Calibration: Level
- Orientation: Orientation setting

The only sensor that needs to be calibrated is the TFMMini Plus. Configure the serial port on which the LiDAR will run by setting the **SENS_TFMINI_CFG** parameter to *Telem3*, which is the port to which this sensor is connected. There is no need to set the baud rate, as it is hardcoded in the sensor driver and only one rate is supported.

3.2 Radio Control Calibration (Futaba T14SG)

Available channels: 14

Used channels: 8



Figure 1: Futaba T14SG Switch Position

Basic control channels:

- Channel 1: Roll
- Channel 2: Pitch
- Channel 3: Throttle
- Channel 4: Yaw

Extended channels:

- Channel 6: Arming Switch (SE)
- Channel 5: Flight mode (SB)

I use 3 different flight modes: Land - Altitude - Position for testing and tuning. And the combination off Land - Altitude - Mission for autonomous flight.

- Channel 7: Loiter (SC)
- Channel 8: Return to Launch (SD)

3.3 Power

Battery type: Lithium-Ion 6S2P Minimum Voltage: 21.6V Maximum Voltage: 24.9V

Encountered Problems: After fully charged the onboard battery indicator show only 50 to 60 percent battery in land mode and when Armed the battery drops to 0, triggering the battery alarm. Even when switch to another fully charged battery it still drop significantly, posing a threat to flight safety.

Solution: Ignore the Battery indicator since the reading is not correct

ESC calibration procedure:

1. Remove the propellers.
2. Disconnect the battery and connect the flight controller via USB (only).
3. Open the QGroundControl Settings > Power, then press the Calibrate button.
4. Connect the battery when prompted:
5. Once the calibration complete you will be prompted to disconnect the battery.

3.4 Actuators

Defining the motor direction and power output/signal input boundaries Figure out the motor's position and change the wiring according to the PX4 Quadcopter Frame X.

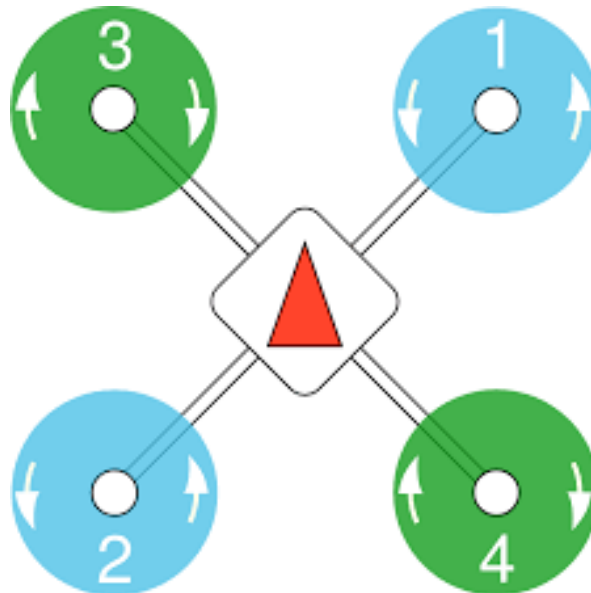


Figure 2: X Frame Motor's Position for PX4

Since the PX4 is in Plug-and-play format, it is easier to use 4 separated ESCs which come with female header instead of using 4-in-1 ESC which require soldering.

3.5 Safety/Failsafe

This section define what to do in worst cases like low battery, lost signal,etc.

- Low Battery: Warning at 15%, 7%, 5%
- Detect Collision: None (No distance sensor for this purpose)
- RC signal lost: Return to launch. However I cannot test this because it only trigger in real flight
- Data link lost (Telemetry): Return to launch
- Return to launch setting: Climb to 10m (avoid all obstacles) - Return to launch position using GNSS data (Straight Path). Land immediately.

3.6 PID Tuning

The QGroundControl PID Tuning setup provides real-time plots of the vehicle setpoint and response curves. The goal of tuning is to set the P/I/D values such that the Response curve matches the Setpoint curve as closely as possible (i.e. a fast response without overshoots). (Source: PX4 | Multicopter PID Tuning Guide)

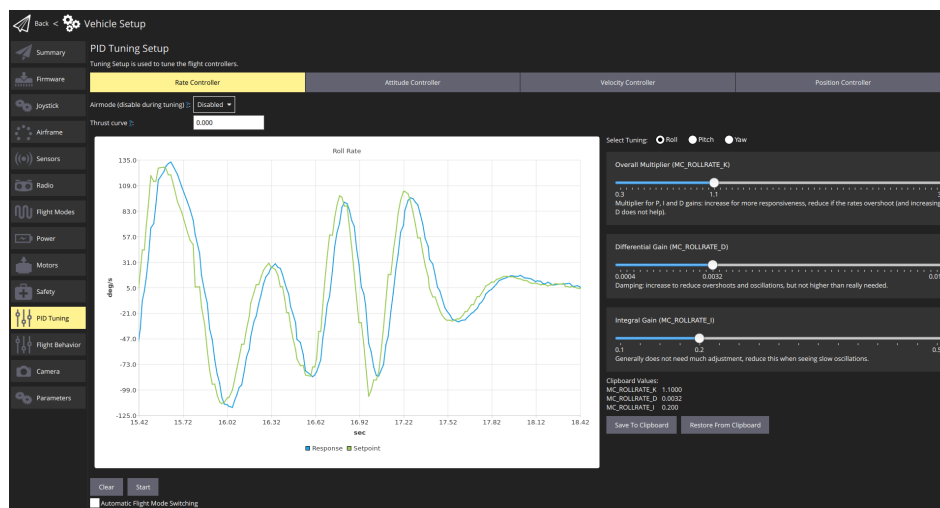


Figure 3: PID tuning ideal case

However, PID control is ideal to be tune with active flying using autotune. Therefore it is best to wait for the maiden flight to apply autotune for PID.

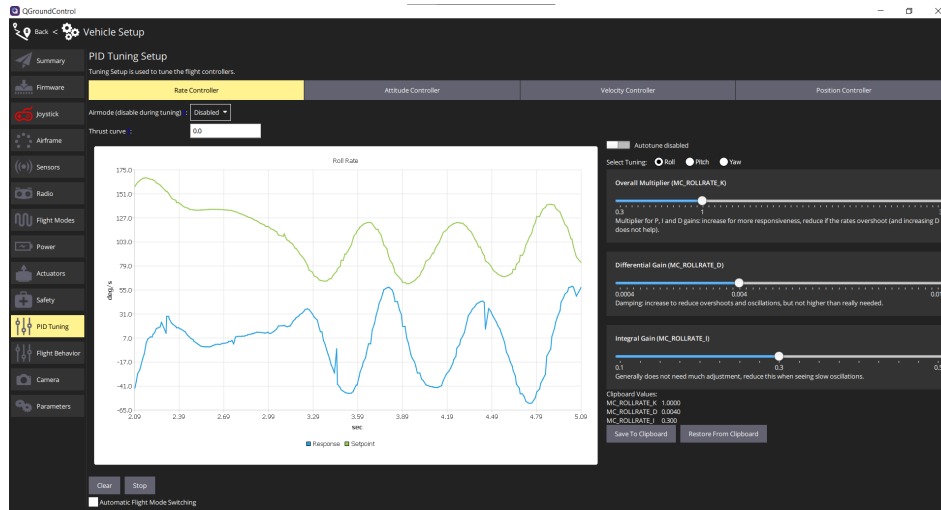


Figure 4: Current PID

4 Mission Planning

Mission planning is a function that provided by autonomous flight controller software like the PX4, which allow the user to preset a mission for the drone by defining parameters like longitude/latitude, altitude, speed, and the action of the drone.

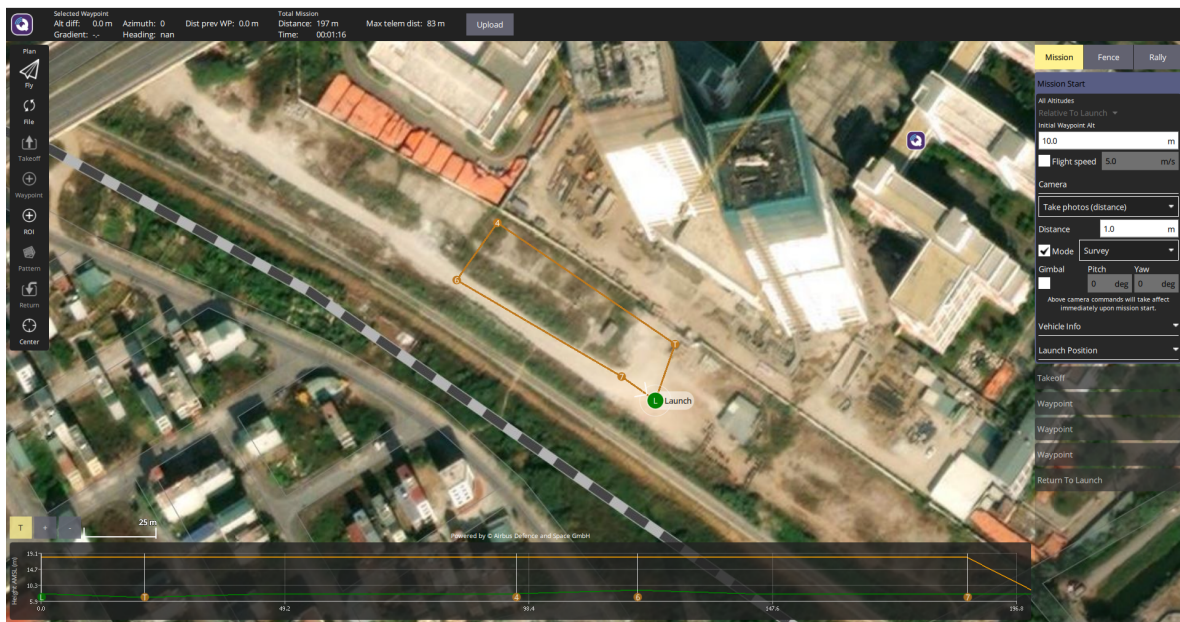


Figure 5: Mission Planning interface

4.1 Create plan

You can create plan from a blank map or use template like Survey, Conrridor scan or Structure scan.

1. **Survey:** This mode is used for capturing large area maps. It automatically generates parallel flight paths over a defined area with the UAV flying back and forth to cover the entire region.

2. **Structure Scan:** Designed for inspecting vertical structures such as buildings or towers, this mode creates a set of waypoints that follow the structure's outline, enabling detailed imaging or measurements of vertical surfaces.
3. **Corridor Scan:** Used for scanning along linear features like roads or power lines, this mode defines a corridor where the UAV will fly back and forth, capturing data along the path.

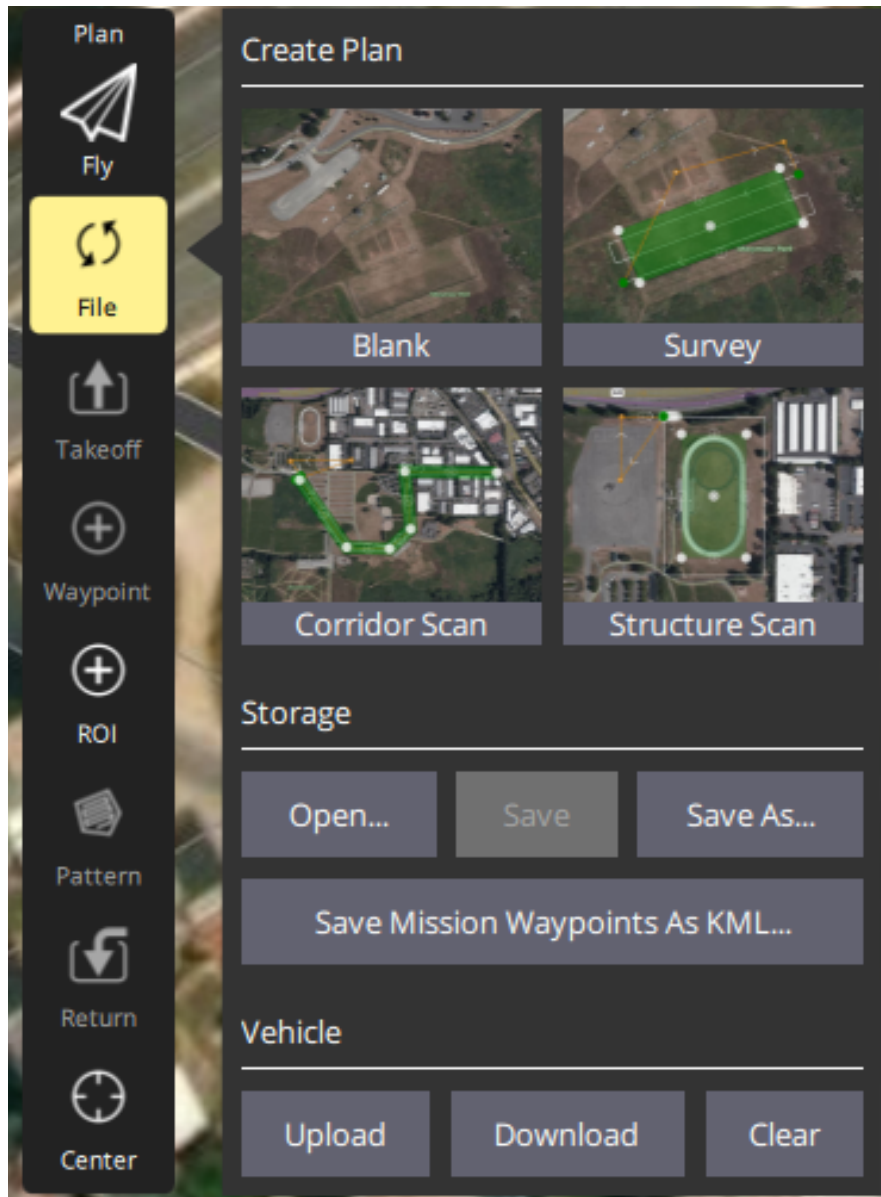


Figure 6: Mission Types

Procedure

1. **Open QGroundControl:** Launch QGroundControl and ensure your UAV is connected.
2. **Navigate to the Mission Planning Interface:** Go to the mission planning screen where you can create and manage mission plans.
3. **Select the Plan Type:** Choose from options like Survey, Structure Scan, or Corridor Scan, or start a *Blank Plan*.

4. Define the Area or Path:

- For Survey: Draw a polygon around the area to be surveyed.
- For Structure Scan: Mark the structure and its height.
- For Corridor Scan: Draw the path along the linear feature.

5. **Set Mission Parameters:** Adjust altitude, speed, overlap, and other parameters.

6. **Add Waypoints (if applicable):** Place waypoints manually or automatically.

7. **Save and Upload:** Save the plan and upload it to your UAV for execution.

4.2 Noticeable reminder

When creating mission plans in QGroundControl, ensure to take note of the following:

1. **Area Coverage:** Ensure the area you define (for Survey, Structure Scan, or Corridor Scan) covers the intended region and that all waypoints are within safe flight limits.
2. **Altitude Settings:** Set appropriate altitudes for different parts of the mission. Ensure the UAV will be at a safe height, avoiding obstacles and complying with regulations.
3. **Overlap and Camera Settings:** For Survey and Structure Scan, set correct overlap (typically 60-80%) for effective imaging and data capture.
4. **Waypoint Placement:** For Corridor and Structure scans, ensure waypoints are placed to cover the entire desired path or structure. Avoid excessive gaps or overlap.
5. **Speed and Time:** Set UAV speed based on the mission type. Slow speeds might be necessary for detailed inspections, while faster speeds are acceptable for large-area surveys.
6. **Battery Life:** Ensure the planned mission fits within the UAV's battery limits. Long missions might need additional planning for battery management or mid-flight charging.
7. **Geofencing:** Enable geofencing to restrict the UAV within a designated area to prevent it from going outside safe boundaries.
8. **Redundancy and Safety Features:** Consider adding return-to-home (RTH) waypoints or other safety measures in case of unexpected events.
9. **Regulatory Compliance:** Make sure the mission plan complies with local UAV regulations, such as altitude limits and restricted zones.

(Source: Missions | PX4 Guide)

Mission
Fence
Rally

Mission Start

All Altitudes
Relative To Launch ▼
Initial Waypoint Alt
10.0 m
☐ Flight speed 5.0 m/s

Camera

Take photos (distance) ▼
Distance 1.0 m
☒ Mode Survey ▼
Gimbal Pitch Yaw
☐ 0 deg 0 deg

Above camera commands will take affect immediately upon mission start.

Vehicle Info ▼
Launch Position ▼

Takeoff
Waypoint
Waypoint
Waypoint
Return To Launch

Figure 7: Mission Steps

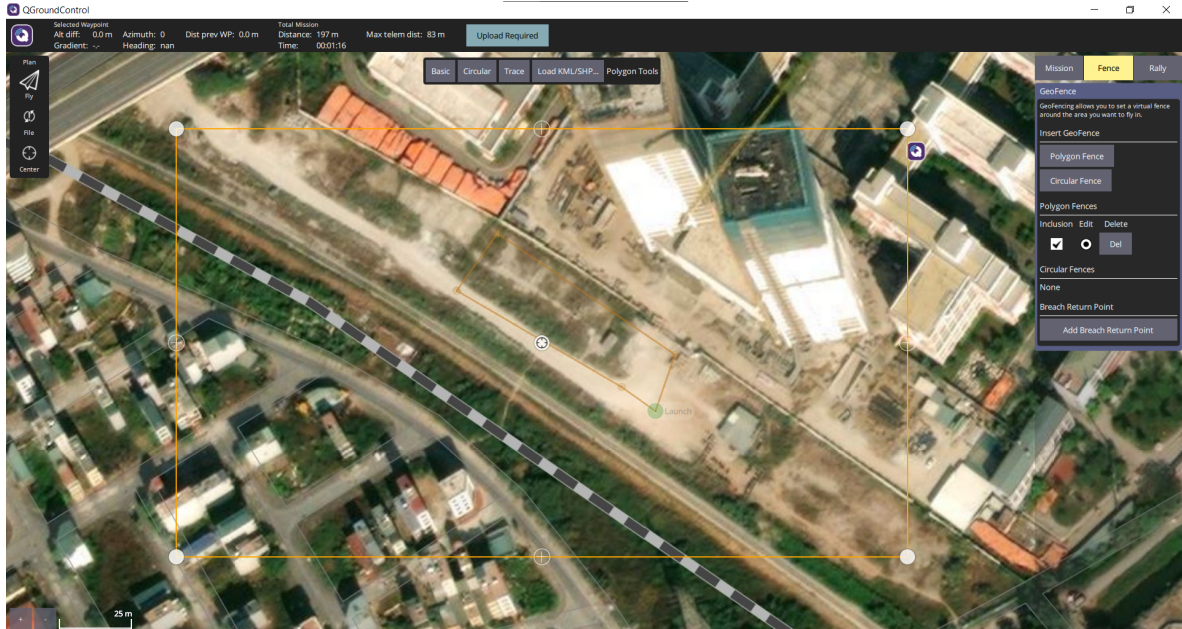


Figure 8: Geo-fence - Flying Zone Boundary

5 Conclusion

The setup and calibration of the Holybro S500 UAV, as outlined in this report, provide a comprehensive guide for preparing the vehicle for safe and effective operation. Key components like the TFMMini Plus sensor, the Futaba T14SG controller, and the PX4 flight controller are carefully configured and calibrated to ensure reliable performance. The detailed procedures for sensor calibration, power management, and ESC setup help resolve common issues, such as battery misreadings and flight controller calibration.

The mission planning section emphasizes the flexibility of QGroundControl in designing customized flight paths for various surveying and scanning tasks, such as Survey, Structure Scan, and Corridor Scan. By following the prescribed steps, users can ensure safe, efficient, and compliant drone operations tailored to their specific needs.

Lastly, it is essential to consider safety protocols, such as battery life management, waypoint placement, geofencing, and regulatory compliance, to enhance the UAV's performance and mitigate potential risks during operation.